Sangam Pratap Singh

sangampratapsingh21012006@gmail.com · +91-9205219678 · random-pikachu.dev · linkedin.com/in/sangam21
github.com/Random-Pikachu · invent.kde.org/randompikachu

Education

Indian Institute of Information Technology and Management, Gwalior Aug 2023 - May 2028
B.Tech in Information Technology (with integrated MBA)

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Machine Learning.

Technical Skills

- **Languages:** C++, C, Go, Python, JavaScript/TypeScript, SQL, Qt/QML
- **Core CS:** Data Structures & Algorithms, Object-Oriented Design, DBMS, Machine Learning, REST APIs
- **Backend & Frameworks:** Node.js, Express.js, Next.js, React.js, WebSockets, PyTorch
- **Databases (SQL & NoSQL):** PostgreSQL, MySQL, Prisma ORM, MongoDB, SpacetimeDB
- **Cloud & DevOps:** AWS (EC2, SES), Azure, Docker, Linux, Git, CI/CD

Experience

Software Engineer Intern (Full Stack) Jun 2026 – Present
Retsguide | TypeScript, Node.js, Next.js, React, PostgreSQL, Prisma, Azure *Remote, Copenhagen*

- **Owned three customer-facing application flows end-to-end** — design through deployment and ongoing operation; designed the **Prisma/PostgreSQL** schema and ran live-data-preserving enum **migrations** on production data with full status-history tracking.
- Built an **AI-assisted document-classification and eligibility pipeline**, defining the extraction and eligibility-decision logic and validating outputs against real case data to catch and correct misclassifications before they reached the case flow.
- Engineered modular **REST services** on a layered **controller–service–repository (object-oriented) design** with **state-machine** status transitions, validated DTOs, idempotent case creation, and PII-sanitized responses.
- **Root-caused** a server-side-rendering bottleneck causing **2s+** page loads to redundant per-request *getServerSideProps* fetches; restructured the data path and cut server response time to **150ms**, a **13× reduction**.

Software Development Intern Jul 2025 – Sep 2025
OpenSphere | Node.js, React.js, WebSockets, AWS SES, MongoDB *Remote, San Francisco*

- Designed and deployed a **concurrent WebSocket service** powering real-time threaded commenting in the platform's text editor; an **event-driven, non-blocking** model served **simultaneous connections** at low latency, driving **2×** editor adoption post-launch.
- Removed a redundant data-filing stage from the **end-to-end drafting workflow**, delivering **2× throughput** and eliminating manual intervention at that step.

Academic Research

Bachelor's Thesis: A Hybrid CNN-Transformer for Building Electricity Load Forecasting Jun – Aug 2026
Python, PyTorch, scikit-learn | CNN-Transformer, CNN-LSTM, CNN-GRU, LSTM, GRU *IIITM Gwalior*

- Proposed and implemented a **CNN-Transformer** — a multi-scale convolutional stem (4 kernel widths), an **FFT-based frequency-domain branch**, and **cross-attention fusion** feeding a Transformer encoder — for hourly load forecasting on the **CU-BEMS** dataset (11,486 hourly records, a 7-story office building, Jul 2018–Oct 2019).
- Benchmarked across a grid of **4 lookback windows** (8–168 h) and forecast horizons of 2–48 h against LSTM, GRU, CNN-LSTM, CNN-GRU and standalone-Transformer baselines; the CNN-Transformer held test $\mathbf{R^2}$ in **0.8816–0.9712** (mean 0.9218) and was the only model that never fell below $R^2 = 0.90$ at a 24 h lookback across every tested horizon.
- Ran a **protocol-matched ablation** isolating the convolutional + spectral front end from the encoder (identical data, loss, optimizer, schedule and seed): the CNN-Transformer beat a standalone Transformer on **15 of 18** lookback–horizon cells ($\Delta R^2 = +0.0136$, –10.4% MAE), with the gain roughly doubling at longer horizons ($H \geq 10$).
- Engineered the full data pipeline — zone-to-floor aggregation, hourly resampling, chronological **70/10/20** split, scalers fit on training data only — and logged **130+** experiment runs (G1–G10) sweeping architecture, feature sets (weather, cyclical encodings, per-floor), lookback and horizon.

Open Source

KDE Plasma Contributor Dec 2025 – May 2026
C++, Qt/QML | Work featured in KDE's This Week in Plasma (Feb 2026) *Remote*

- **Wallpaper Gallery Caching (MR 6280):** Added local caching to the Plasma Wallpaper Gallery with **SHA-256** content hashing for deduplication, keeping entries intact when source files move or are deleted.
- **Wallpaper Path Persistence (MR 6208):** Eliminated repeated directory-browsing friction by persisting picker state via **KConfig**; validated through **functional testing** across distributions and merged after **maintainer code review**; now shipping to **millions of Plasma users**.
- Additional merged MRs: WCAG keyboard accessibility in Plasma Digital Clock, todo state-inheritance bug fix in Marknote, and spacebar canvas pan in Drawy.

Projects

Astra | *Electron, Node.js, Express, MongoDB, Tree-sitter, Gemini AI* Aug – Sep 2025
https://github.com/Random-Pikachu/Astra

- Architected a **GenAI developer-productivity tool**: a **static-analysis pipeline** pairing **Tree-sitter AST parsing** with **Gemini** to surface CVEs via **OSV.dev** and auto-generate remediation patches across **500+ files**, delivered as programmatic **pull requests** — zero manual steps from detection to code review.

CodeRivals | *React.js, SpacetimeDB, AWS EC2, Docker, Nginx, Monaco Editor* Apr 2026
https://github.com/Random-Pikachu/code-rivals

- Developed a **real-time 1v1 competitive coding platform** where two players solve the same problem live; implemented the matchmaking queue, live match state sync, and an **Elo rating** update after each match.
- Self-hosted the open-source **Piston** execution engine on **AWS EC2** behind an **Nginx** reverse proxy, so submitted code runs inside a **sandboxed Docker container** instead of on the app server; wrote a **C++** parser that converts raw test-case input into the argument types each problem expects.

DevTrackr | *Go, PostgreSQL, SQL, React.js, Docker, Resend* Apr 2026
https://github.com/Random-Pikachu/DevTrackr-Backend

- Created a **scalable Go aggregation service** over **PostgreSQL** unifying the **GitHub, LeetCode, and Codeforces** APIs into one normalized schema with **optimized SQL queries**; added a scheduled digest engine dispatching **HTML email** summaries. Released to real users.

Achievements

- Rated **1799** on **LeetCode** and **1329 (Pupil)** on **Codeforces** across **12** rated contests, reflecting sustained algorithmic problem-solving practice.
- Reached the **finalist** round at **Hackbyte**, a competitive hackathon, out of a large applicant pool.
- Completed **Sekiro: Shadows Die Twice** on **5** separate playthroughs - a notoriously punishing action title (GOTY 2019) that rewards patience, pattern-reading, and repetition under pressure.